D2L Simulator Mobile Application

Kennesaw State University

CS 4322: Mobile Software Development

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Group Members:

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Acknowledgments

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Abstract

The D2L Simulator Mobile Application is an Android-based application designed to replicate the key functionalities of the Desire2Learn (D2L) learning platform. The purpose of this project is to provide students with an accessible and intuitive interface to navigate course materials such as the syllabus, lab instructions, project guidelines, and the learning modules.

The application was developed using Android Studio and Jetpack Compose, with a focus on modern UI design principles and state-driven navigation. Core features include a login interface, a home navigation system, a "Start Here" section containing essential course documents, and a "Learning Modules" section that allows users to browse the class's modules and access slides and lab resources. Slides are stored locally within the application using the assets directory, while lab materials are accessed via external web links.

The development process involved designing multiple composable UI components, implementing navigation logic using state management, integrating local file handling through Android's FileProvider, and enabling external intents for web-based content.

Results demonstrate that the application successfully mimics the structure and navigation flow of a learning management system. It provides users with a seamless experience for accessing the Mobile Software Development's course content.

Future improvements include enhanced authentication, real-time data integration for registered users, and improved UI refinement. Overall, the project demonstrates practical application of mobile development concepts and modern Android design practices. Outside of this projects scope, but would improve the application beyond its initial purpose would be for this application to store all of a users course material. Allowing users to create their own paths and directories and effectively turn this D2L representation into a record for all

of users college courses.

1 Introduction

This project aims to develop a simplified mobile application that replicates the core structure of Kennesaw State Universities' D2L course interface. The application allows users to log in, navigate essential course resources, and access learning modules in a streamlined manner.

The main objectives of this project are:

- Design a simple, user-friendly mobile interface
- Implement structured navigation
- Provide access to downloadable course materials
- Integrate local and web-based resources

2 Literature Review

Modern Android development has shifted toward declarative UI frameworks such as Jetpack Compose, which simplifies UI creation and improves maintainability. Compose enables developers to build responsive interfaces using composable functions and state-driven design.

Key advantages include:

- Declarative UI over imperative UI
- Efficient state management
- Modular component-based design
- Seamless integration with Android intents

Additionally, Android's FileProvider system allows secure access to local files, making it possible to distribute educational resources directly within applications.

3 Methods

The application was developed using Android Studio and Jetpack Compose.

3.1 Application Structure

- MainActivity – Entry point of application
- Navigation system using state variables
- Multiple composable screens

3.2 UI Components

- Column, Row, LazyColumn layouts
- OutlinedButton and TextButton components
- Custom Header composable

3.3 File Handling

- Assets folder for PDF storage
- FileProvider for secure access
- Helper functions:
  - openAssetFile()
  - openLink()

3.4 External Integration

- Lab materials accessed via web links
- Slides accessed via local PDFs

4 Results

The application successfully implements most planned features and navigation flows. The one missing part of the application is the database for registering users. While the login screen allows users to enter their email and password, currently there is no way for users to actually "register" and be remembered. To bypass this, the sign in button functions to move users onto the next screen as if the user has already registered and correctly entered their credentials.

4.1 Application Screens

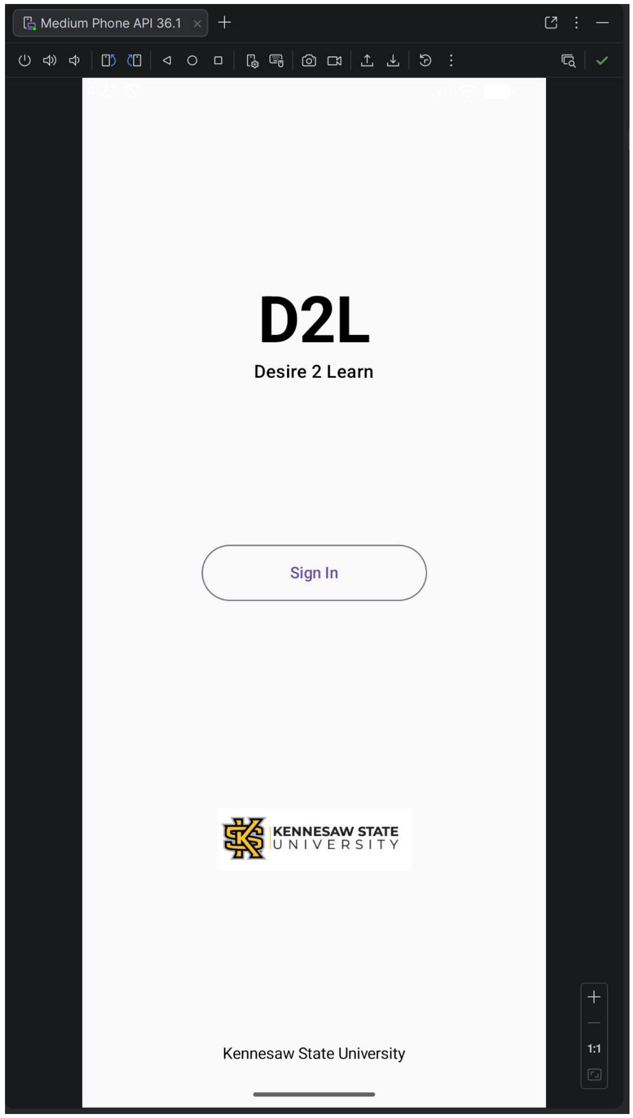

Figure 1: Login Screen

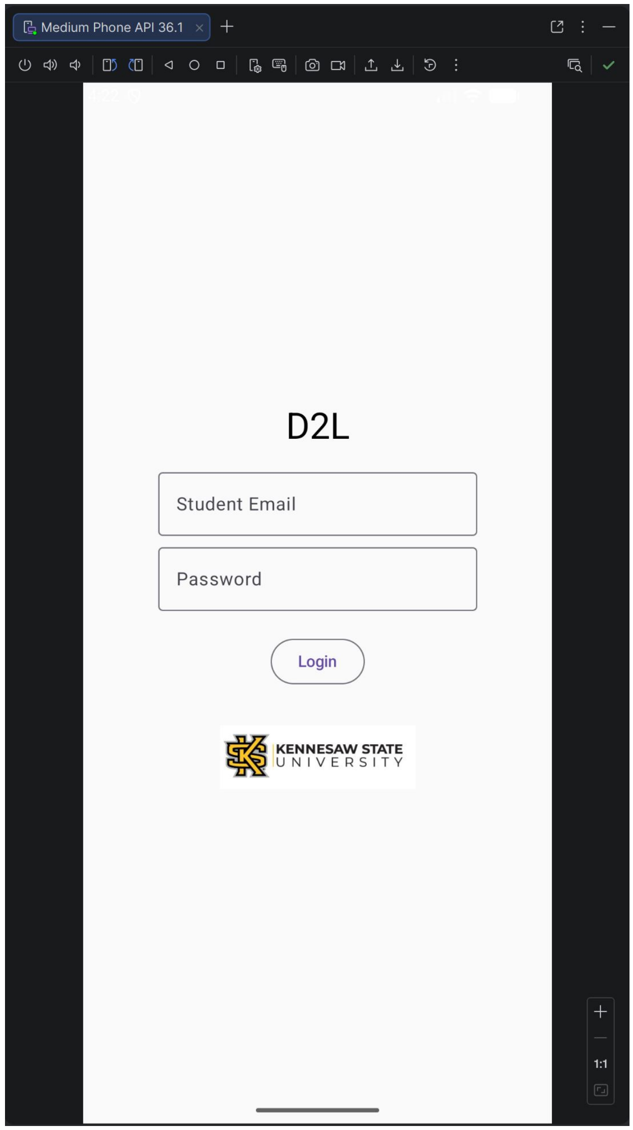

Figure 2: Credential Screen

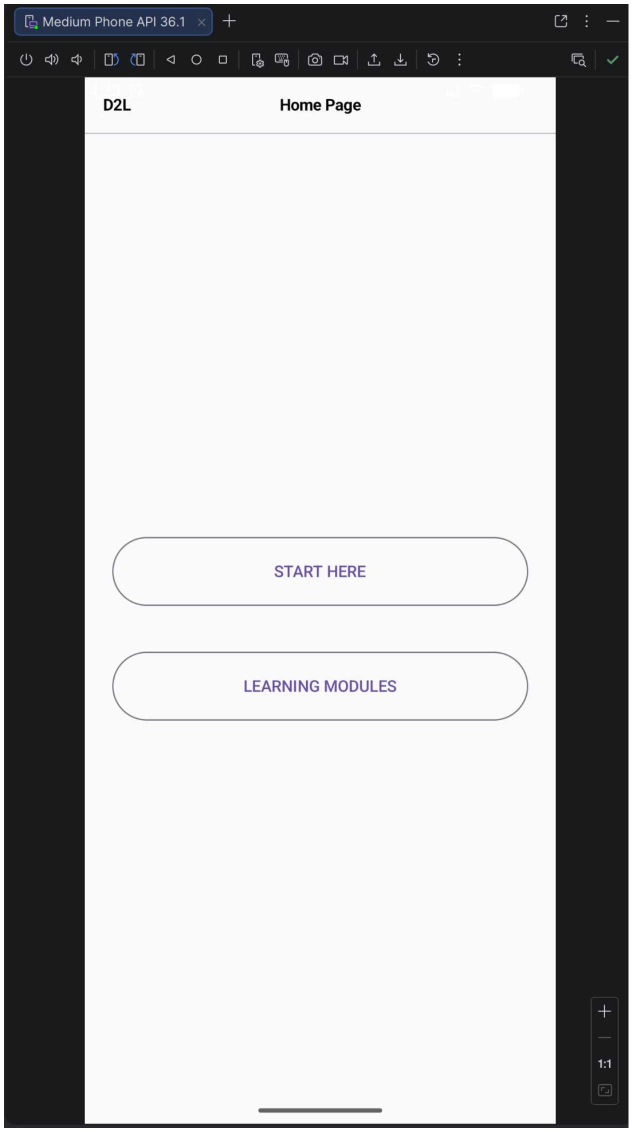

Figure 3: Home Screen

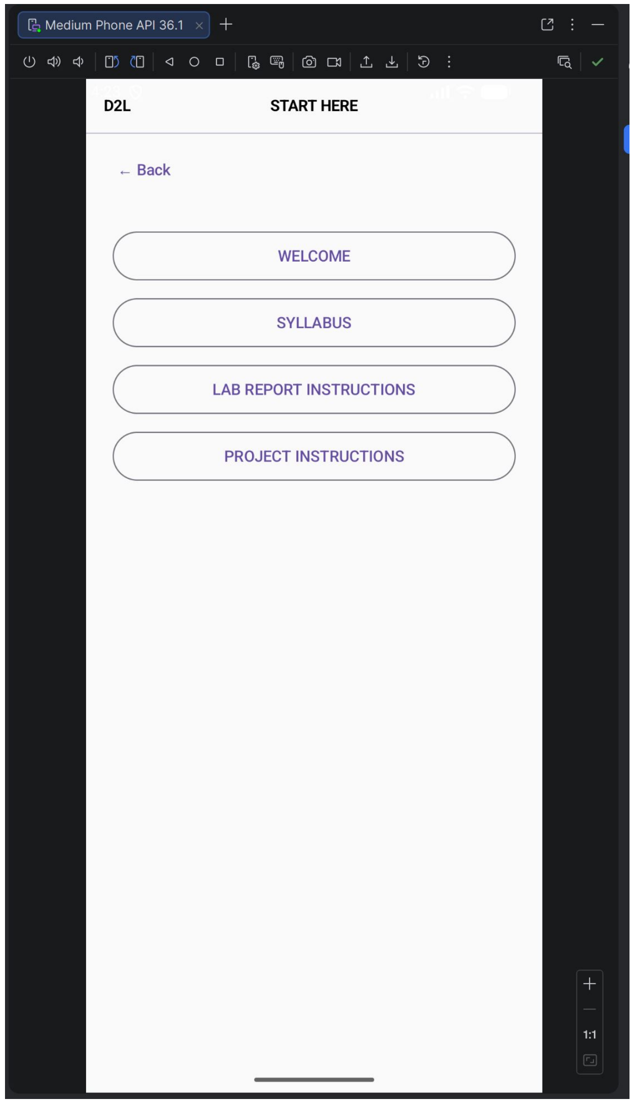

Figure 4: Start Here Screen

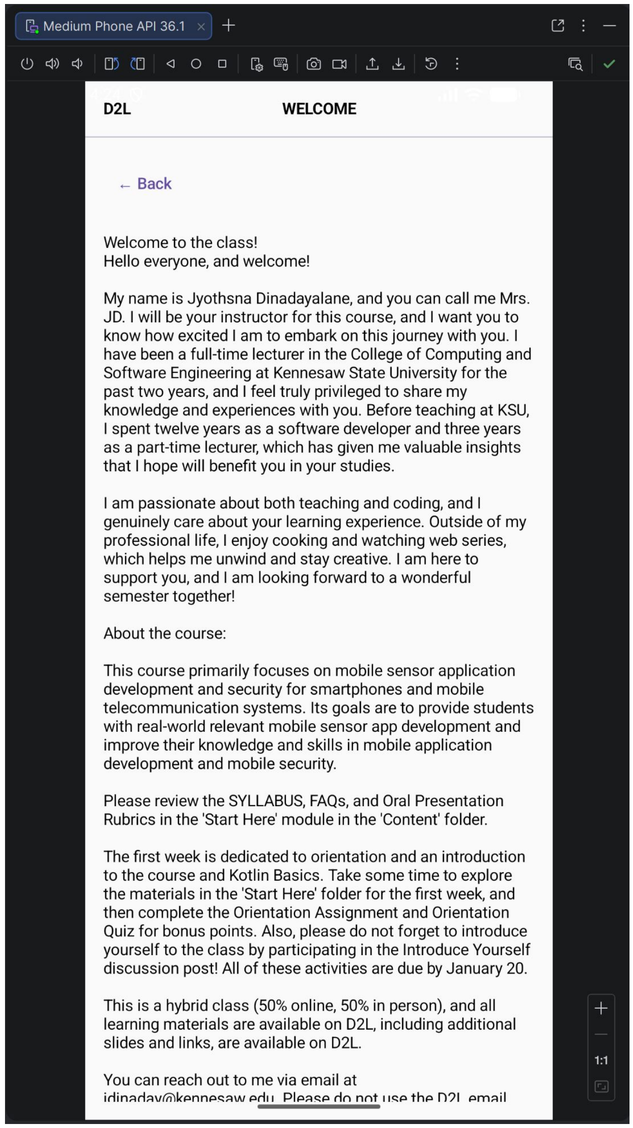

Figure 5: Welcome Screen

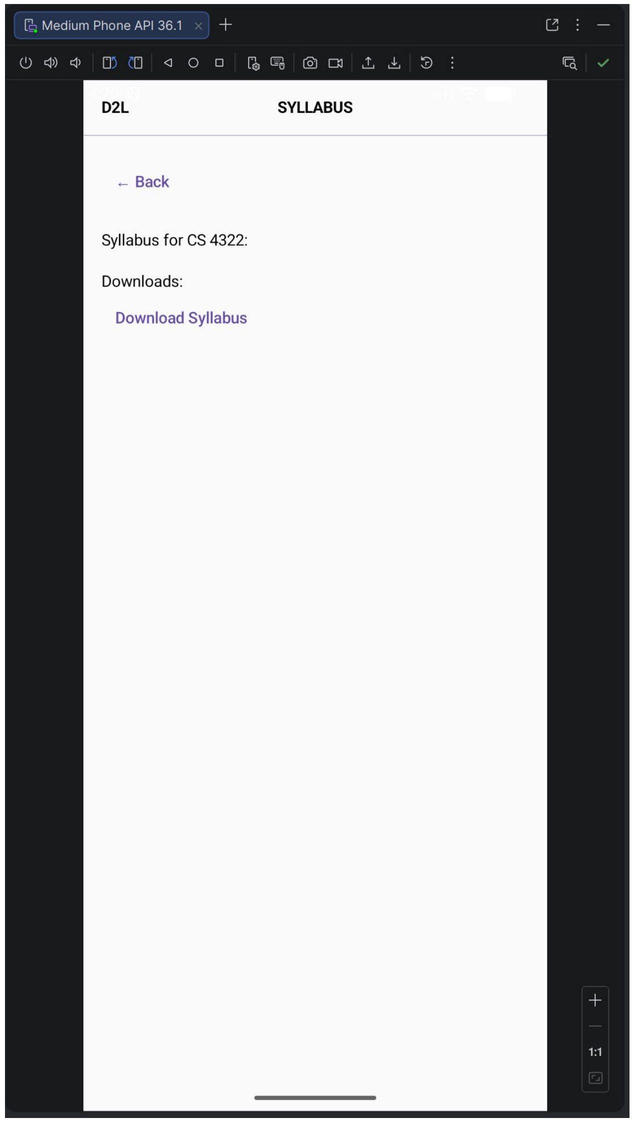

Figure 6: Syllabus Screen

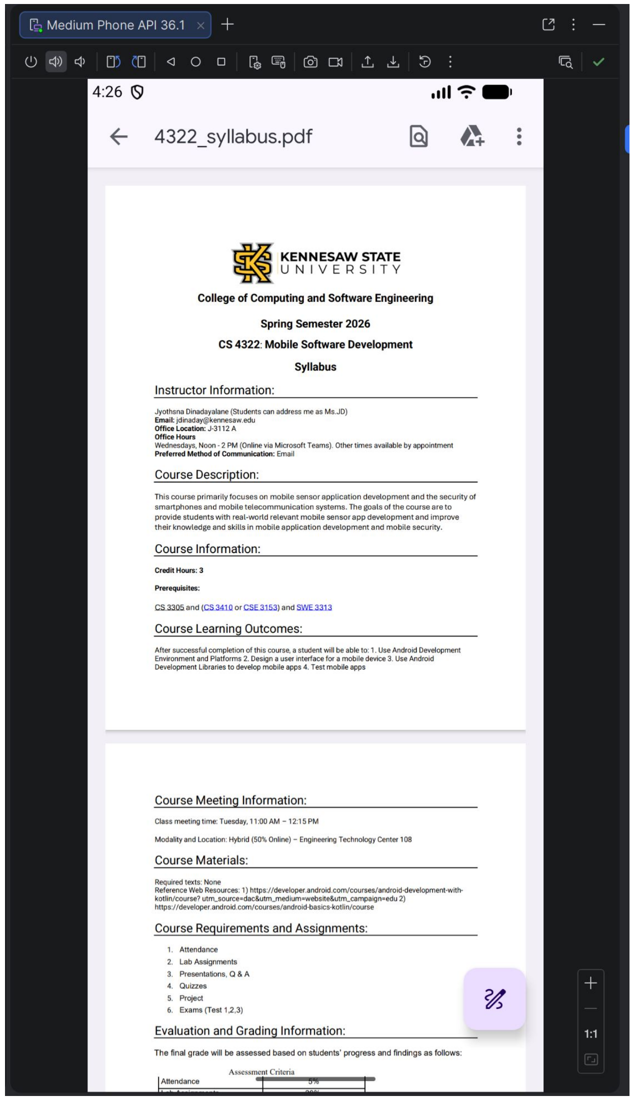

Figure 7: Downloadable Syllabus

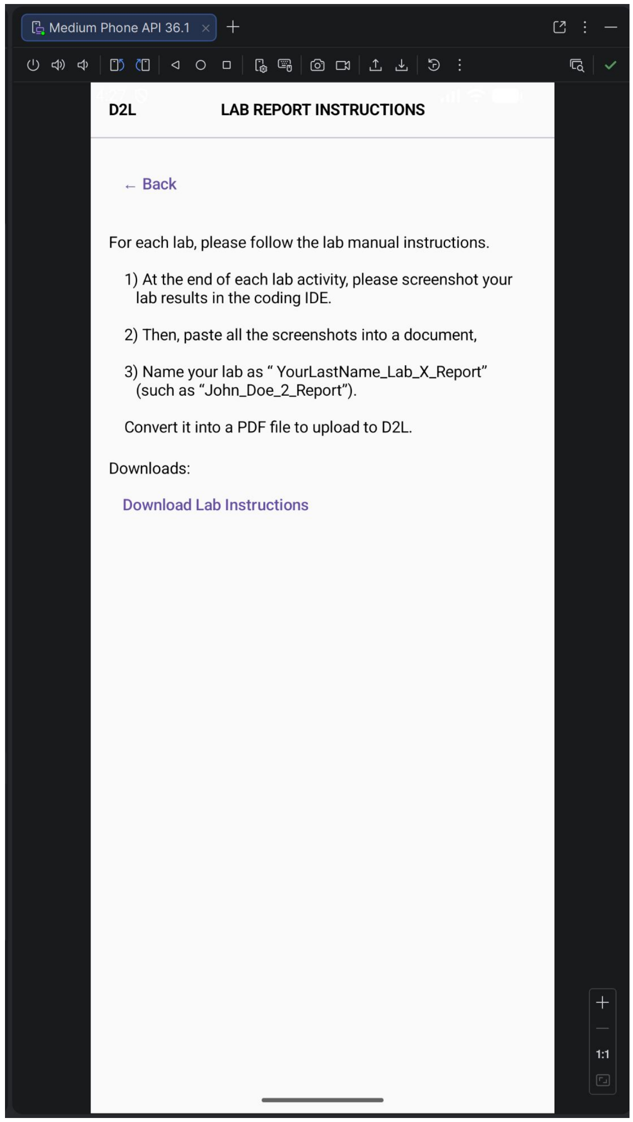

Figure 8: Lab Report Screen

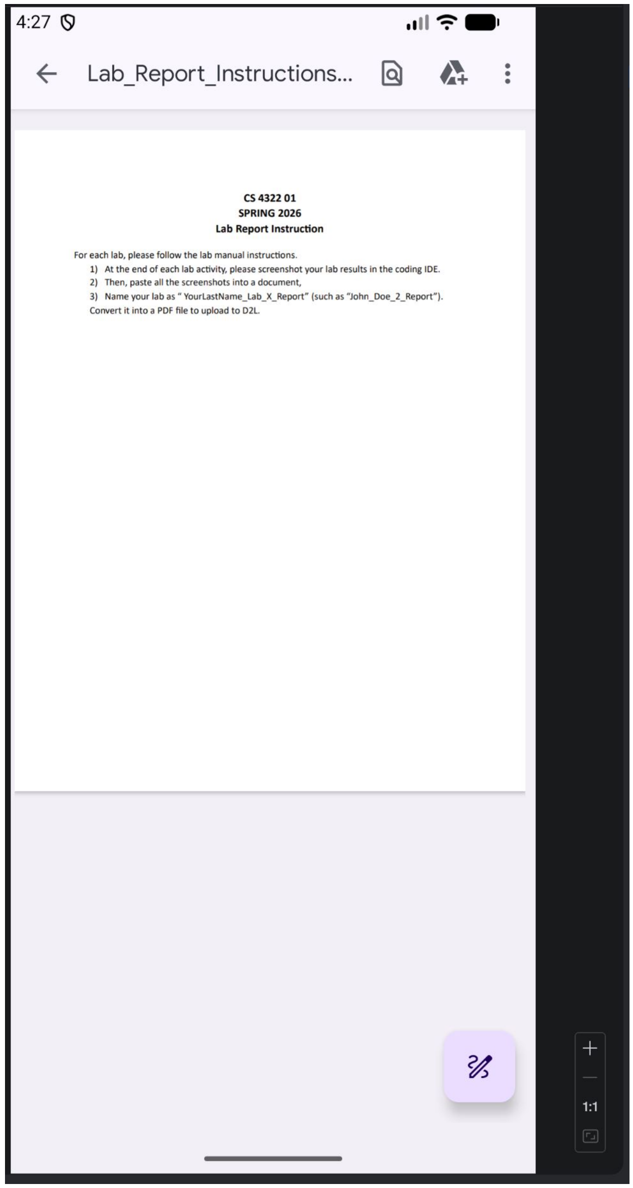

Figure 9: Downloadable Lab Report

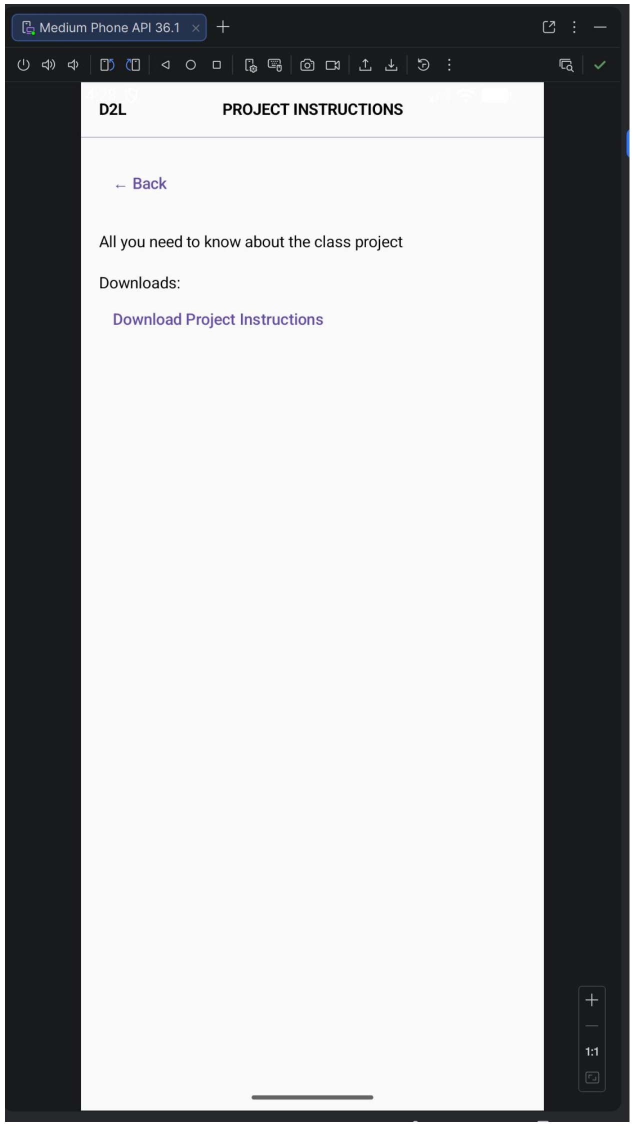

Figure 10: Project Screen

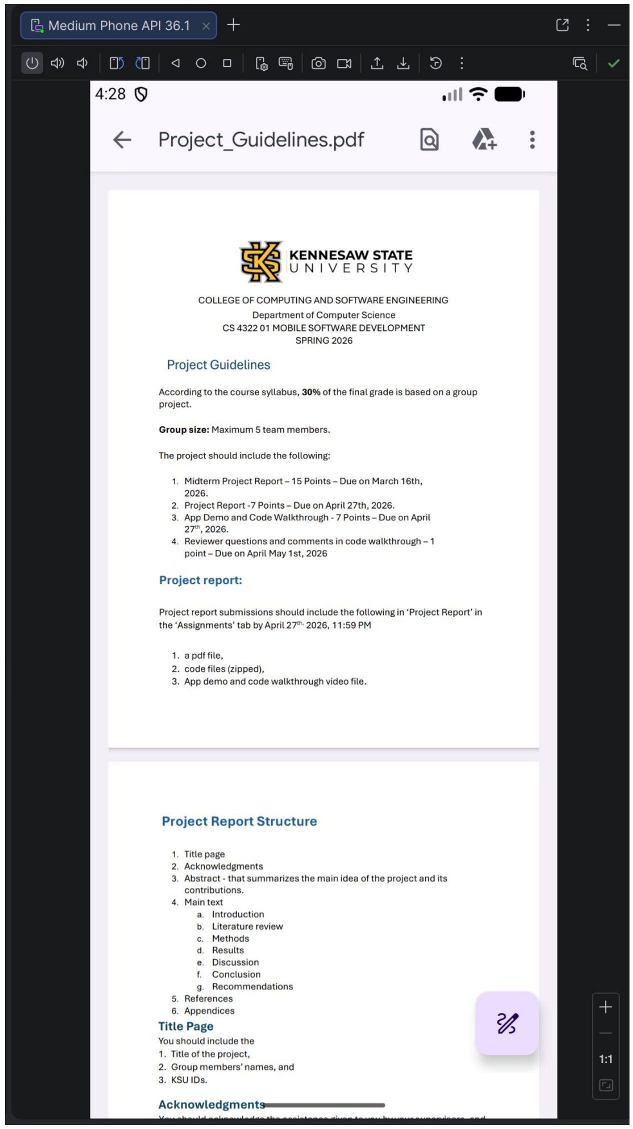

Figure 11: Downloadable Project Instructions

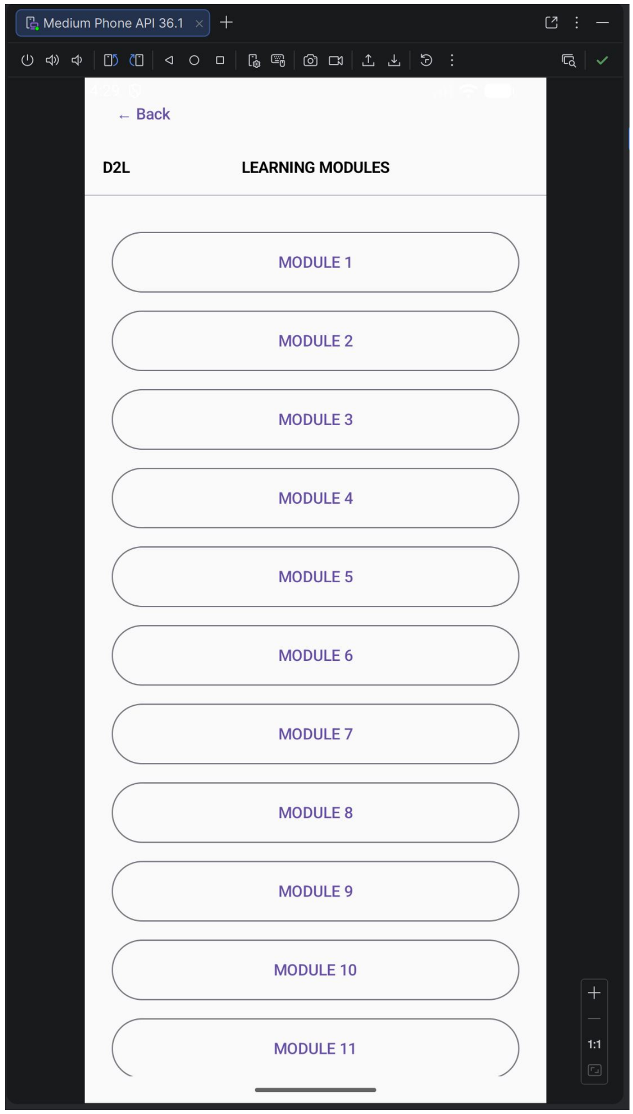

Figure 12: Modules Screen

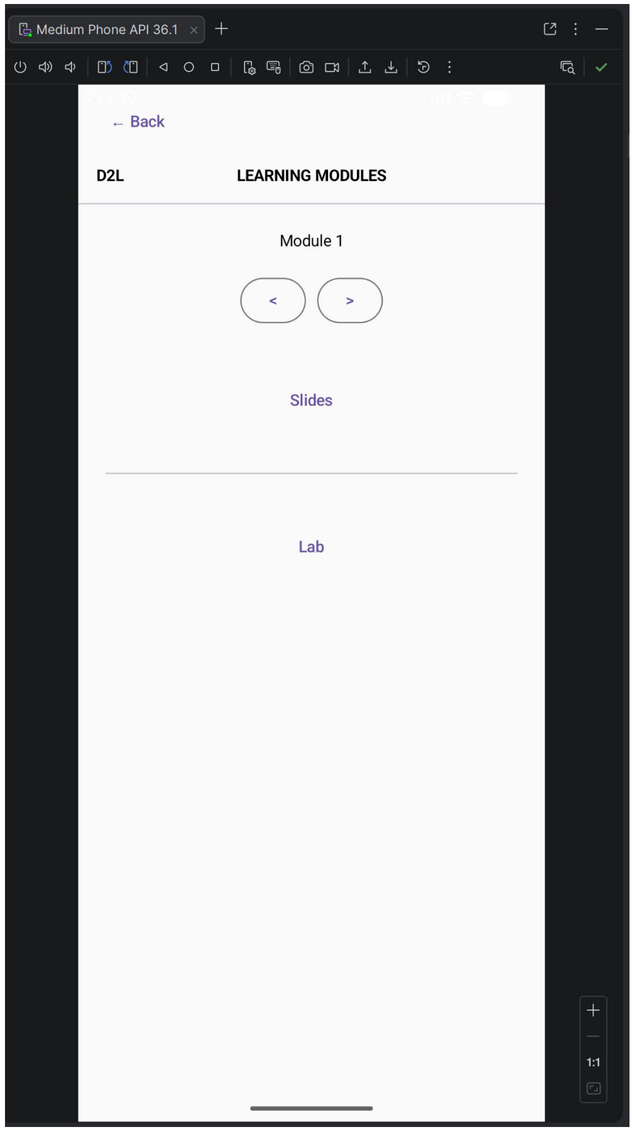

Figure 13: Selected Module Screen

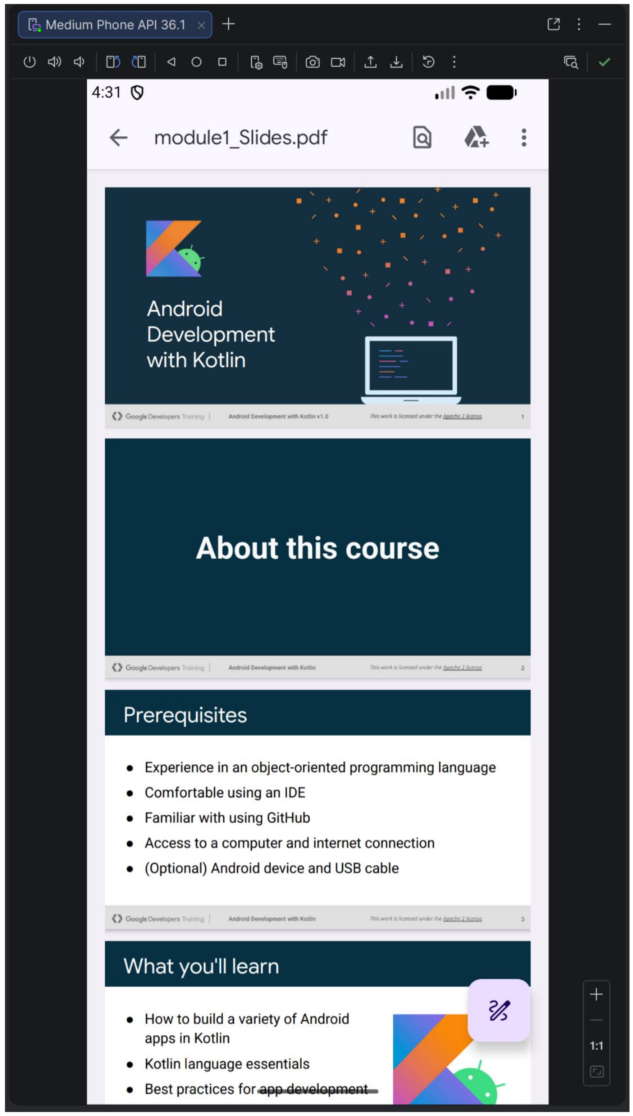

Figure 14: Selected Module's Slides

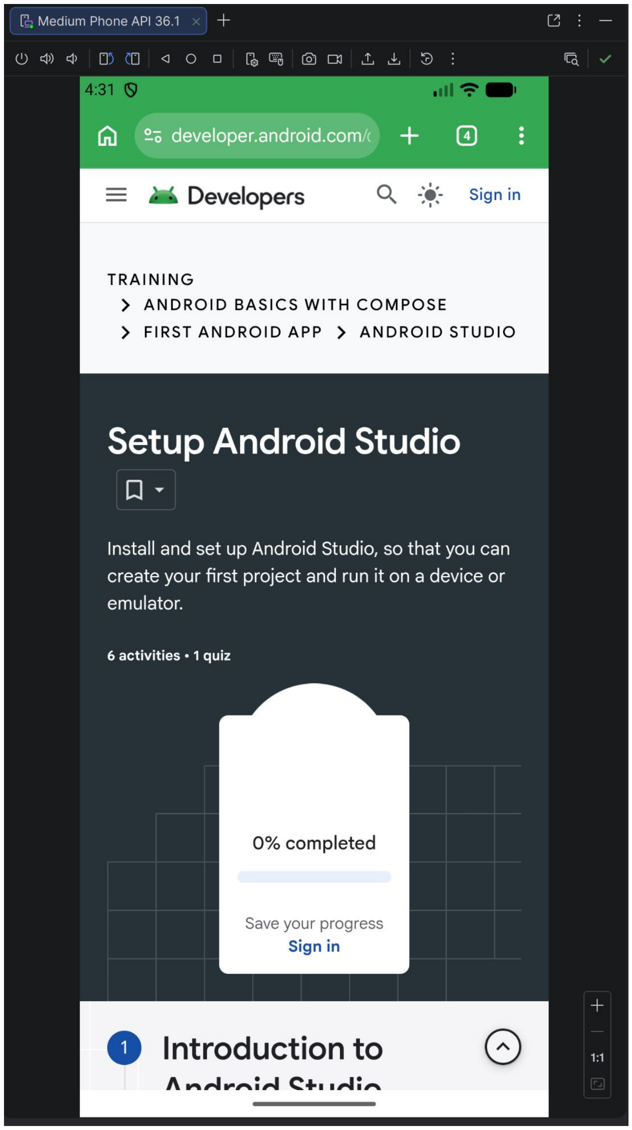

Figure 15: Selected Module's Lab

4.2 Navigation Flow

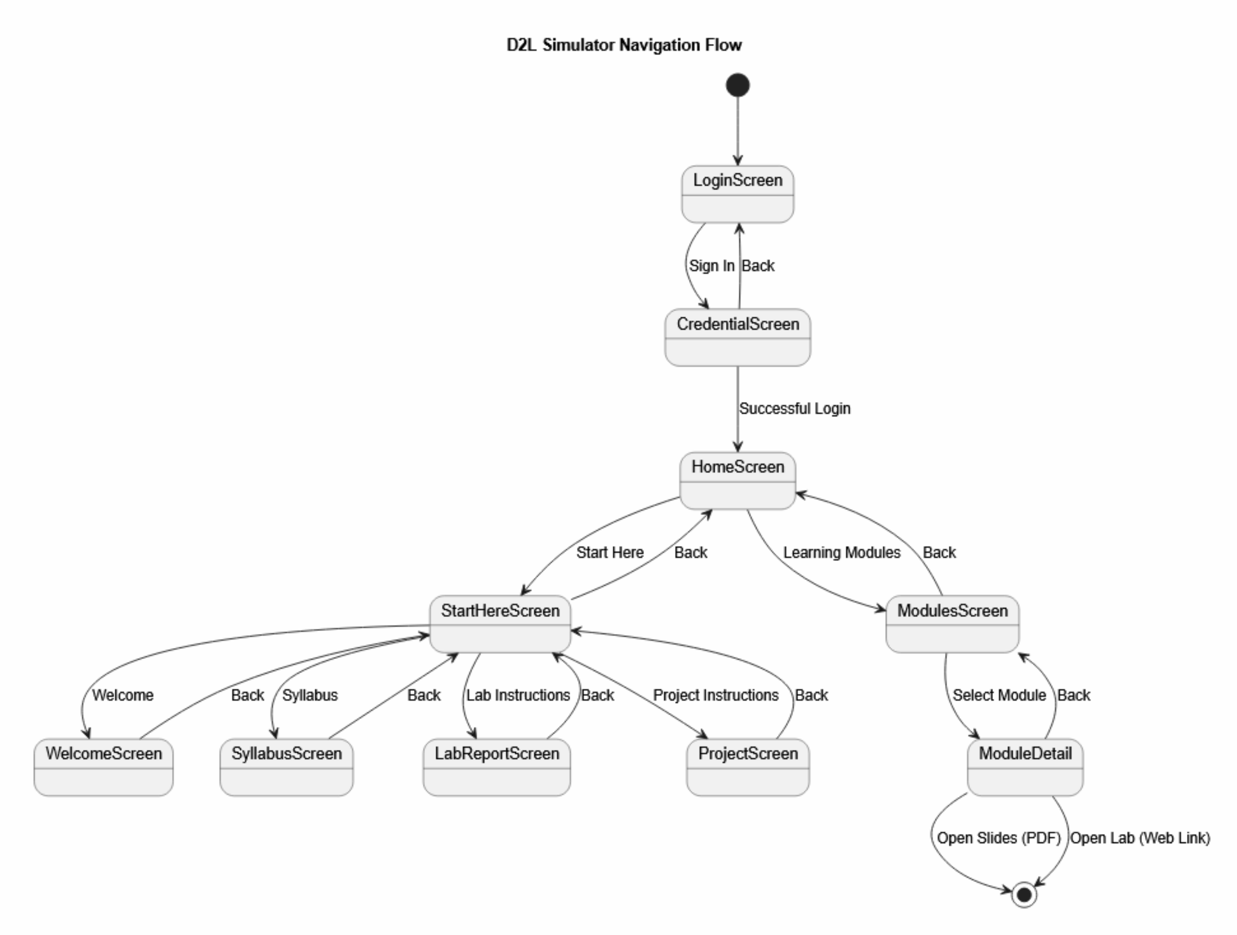

Figure 16: Navigation Flow

4.3 Key Outcomes

- Successful navigation between screens
- Scrollable module list
- Functional file downloads
- External link integration

5 Discussion

The project demonstrates the effectiveness of Jetpack Compose in building modern Android applications.

5.1 Strengths

- Clean UI design
- Functional navigation
- Integration of local and external resources

5.2 Limitations

- No authentication system
- Static content, nothing can be changed within the app itself
- Limited error handling

6 Conclusion

The D2L Simulator application successfully achieves its objective of providing a simplified mobile interface for accessing course materials. The implementation demonstrates key mobile development concepts including UI design, navigation, and file handling.

7 Recommendations

Future improvements include:

- User authentication
- Backend integration
- Improved UI design

- Embedded PDF viewer
- Dynamic content, add/remove material
- Allow users to upload completed Assignments and Labs for later review

References

- Android Developers Documentation: `https://developer.android.com`
- Jetpack Compose Guide
- Course materials and lecture notes

Appendices

- Source Code (Zipped)
- App Demo Video
- Code Walkthrough Video